

Numerical Integration

Newton-Cotes Formula and Gauss Quadrature Formula

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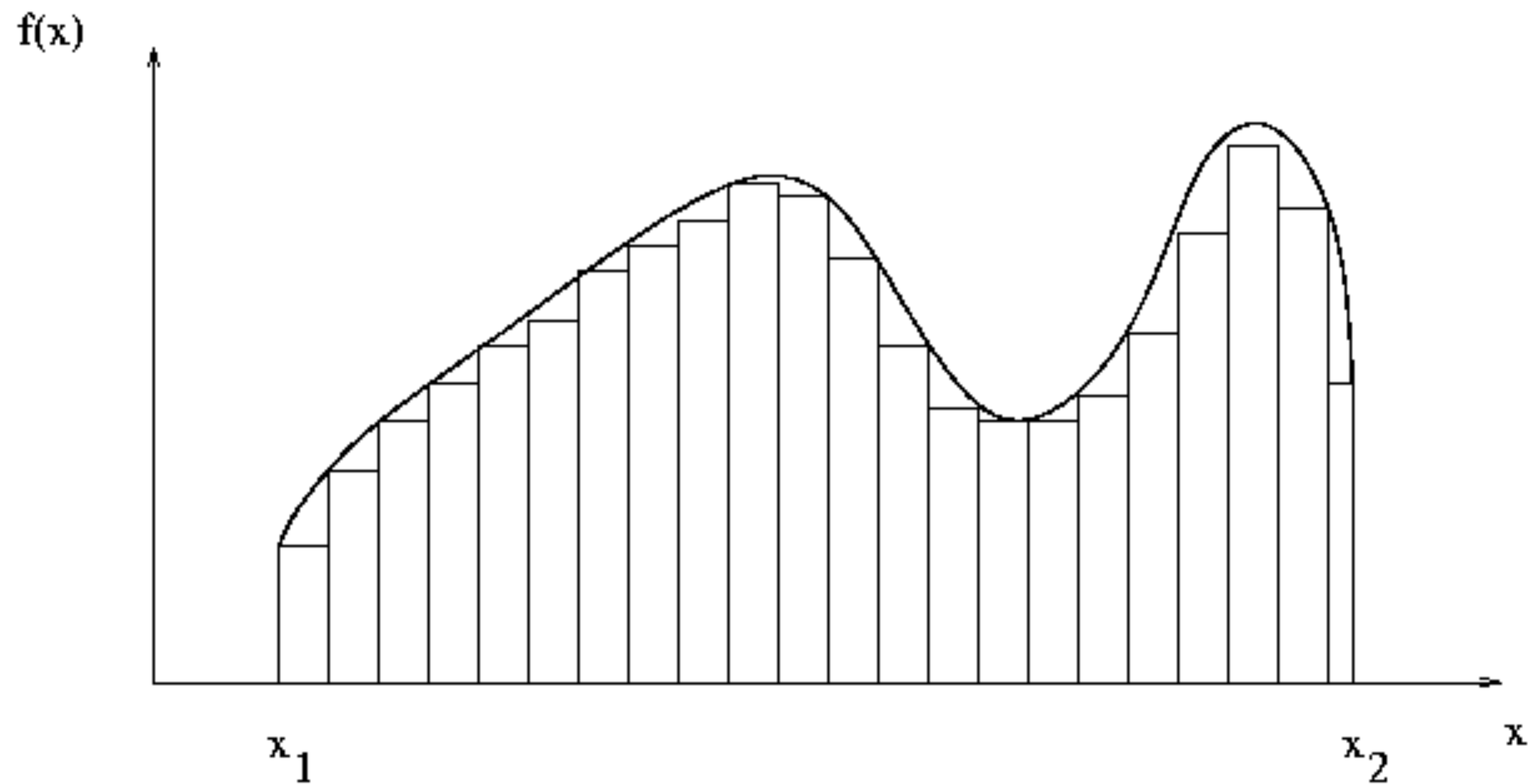
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Introduction

What is Numerical Integration?

- Process of approximating definite integrals
- Useful when:
 - Function is too complex to integrate analytically
 - No closed-form antiderivative exists

What's the big idea behind this?



Newton-Cotes Formulas

Newton-Cotes Integration Overview

- Based on polynomial interpolation over equally spaced points
- **Closed** (includes endpoints): Trapezoidal, Simpson's rule.
- **Open** (excludes endpoints): Midpoint rule, etc.
- We are approximating the definite integral like this:

- $$\int_a^b f(x) dx \approx \sum_{i=0}^n w_i f(x_i)$$

How we get the expression for Newton-Cotes Formula?

- $f(x) \approx f(a) + p\Delta f(a) + \frac{p(p-1)}{2!}\Delta^2 f(a) + \dots + \frac{p(p-1)\dots(p-n+1)}{n!}\Delta^n f(a)$

- Where:

$$x = a + ph, \text{ i.e., } p = \frac{x-a}{h}$$

- So, $I = \int_a^b f(x) dx$

As, $x = a + ph \Rightarrow dx = hdp$; and when $x = a$ then $p = 0$ and when $x = b = x_n$ then,

$$p = \frac{x_n - a}{h} \Rightarrow \frac{a + nh - a}{nh} \quad [\because x_0 = a, x_1 = a + h, x_2 = a + 2h \dots x_n = a + nh]$$

$$= \int_0^n f(a + ph) h dp$$

$$\approx h \int_0^n \left[f(a) + p\Delta f(a) + \frac{p(p-1)}{2!}\Delta^2 f(a) + \dots \right] dp$$

How we get the expression for Newton-Cotes Formula? (Contd..)

$$\begin{aligned}
 \bullet \int_a^b f(x) dx &= h \int_0^n \left[f(a) + p \Delta f(a) + \frac{p(p-1)}{2!} \Delta^2 f(a) + \frac{p(p-1)(p-2)}{3!} \Delta^3 f(a) + \dots \right] dp \\
 &= h \left[f(a) \int_0^n dp + \Delta f(a) \int_0^n p dp + \Delta^2 f(a) \int_0^n \frac{p(p-1)}{2!} dp + \Delta^3 f(a) \int_0^n \frac{p(p-1)(p-2)}{3!} dp + \dots \right] \\
 &= h \left[n f(a) + \frac{n^2}{2} \frac{\Delta f(a)}{1!} + \left(\frac{n^3}{3} - \frac{n^2}{2} \right) \frac{\Delta^2 f(a)}{2!} + \left(\frac{n^4}{4} - n^3 + n^2 \right) \frac{\Delta^3 f(a)}{3!} + \dots \right]
 \end{aligned}$$

This is our Newton-Cotes Quadrature formula. When we choose different numbers of segments (like $n=1$ or $n=2$), we get specific methods like the Trapezoidal Rule and Simpson's Rule.

Simple Trapezoidal Rule (n=1; two points)

$$\bullet \quad I = h \left[y_0 + \frac{\Delta y_0}{1!} \frac{1}{2} \right]$$

$$= h \left[y_0 + \frac{y_1 - y_0}{2} \right]$$

$$= h \left[\frac{2y_0 + y_1 - y_0}{2} \right]$$

$$= \frac{h}{2} [y_0 + y_1]$$

Composite Trapezoidal Rule

$$\begin{aligned} \bullet \quad I &= \int_{a=x_0}^{b=x_n} f(x)dx = \int_{x_0}^{x_1} f(x)dx + \int_{x_1}^{x_2} f(x)dx + \int_{x_2}^{x_3} f(x)dx + \cdots + \int_{x_{n-1}}^{x_n} f(x)dx \\ &= \frac{h}{2} [y_0 + y_1] + \frac{h}{2} [y_1 + y_2] + \frac{h}{2} [y_2 + y_3] + \cdots + \frac{h}{2} [y_{n-1} + y_n] \\ &= \frac{h}{2} [y_0 + y_n + 2(y_1 + y_2 + \cdots + y_{n-1})] \\ \therefore I &= \frac{h}{2} \left[y_0 + y_n + 2 \sum_{i=1}^{n-1} y_i \right] \end{aligned}$$

Sample Problem: Evaluate - $\int_0^1 e^x dx$

Trapezoidal Rule (n=1)

- Divide the interval $[0, 1]$ into one segment with points: $x_0 = 0, \quad x_1 = 1$
- Step size: $h = 1 - 0 = 1$
- Apply Trapezoidal formula:

$$I \approx \frac{h}{2} [f(x_0) + f(x_1)] = \frac{1}{2} [e^0 + e^1] = \frac{1}{2}(1 + e)$$

$$I \approx \frac{1 + 2.71828}{2} = 1.85914$$

Simple Simpson's 1/3 Rule (n=2; three points)

$$\begin{aligned} \bullet \quad I &= h \left[2y_0 + \frac{\Delta y_0}{1!} \cdot \frac{2^2}{2} + \frac{\Delta^2 y_0}{2!} \cdot \left(\frac{2^3}{3} - \frac{2^2}{2} \right) + \frac{\Delta^3 y_0}{3!} \left(\frac{2^4}{4} - 2^3 + 2^2 \right) + \dots \right] \\ &= h \left[2y_0 + \frac{\Delta y_0}{1!} \cdot 2 + \frac{\Delta^2 y_0}{2!} \left(\frac{8}{3} - 2 \right) \right] \\ &= h \left[2y_0 + 2(y_1 - y_0) + \frac{y_2 - 2y_1 + y_0}{3} \right] \quad (\because \Delta^2 y_0 = \Delta y_1 - \Delta y_0 = y_2 - y_1 - (y_1 - y_0) = y_2 - 2y_1 + y_0) \\ &= \frac{h}{3} [6y_0 + 6y_1 - 6y_0 + y_2 - 2y_1 + y_0] \\ \therefore I &= \frac{h}{3} [y_0 + 4y_1 + y_2] \end{aligned}$$

Composite Simpson 1/3 Rule

$$\begin{aligned} \bullet \quad I &= \int_{a=x_0}^{b=x_n} f(x)dx = \int_{x_0}^{x_2} f(x)dx + \int_{x_2}^{x_4} f(x)dx + \int_{x_4}^{x_6} f(x)dx + \cdots + \int_{x_{n-2}}^{x_n} f(x)dx \\ &= \frac{h}{3} [y_0 + 4y_1 + y_2] + \frac{h}{3} [y_2 + 4y_3 + y_4] + \frac{h}{3} [y_4 + 4y_5 + y_6] + \cdots + \frac{h}{3} [y_{n-2} + 4y_{n-1} + y_n] \\ &= \frac{h}{3} \left[y_0 + y_n + 2 \sum_{i=1, i=\text{even}}^{n-1} y_i + 4 \sum_{i=1, i=\text{odd}}^{n-1} y_i \right] \end{aligned}$$

Sample Problem: Evaluate - $\int_0^1 e^x dx$

Simpson's 1/3 Rule (n=2)

- Step size: $h = \frac{1 - 0}{2} = 0.5$
- Divide the interval $[0, 1]$ into two equal subintervals: $x_0 = 0$, $x_1 = 0.5$, $x_2 = 1$
- Apply Simpson's formula:

$$\begin{aligned} I &\approx \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] = \frac{0.5}{3} [e^0 + 4e^{0.5} + e^1] \\ &= \frac{0.5}{3} [1 + 4(1.64872) + 2.71828] = \frac{0.5}{3} (1 + 6.59488 + 2.71828) = \frac{0.5}{3} (10.31316) \\ &= 0.1667 \times 10.31316 = 1.71886 \end{aligned}$$

Simple Simpson's 3/8 Rule (n=3; 4 points)

$$\begin{aligned} \bullet \quad I &= h \left[3y_0 + \frac{\Delta y_0}{1!} \cdot \frac{3^2}{2} + \frac{\Delta^2 y_0}{2!} \cdot \left(\frac{3^3}{3} - \frac{3^2}{2} \right) + \frac{\Delta^3 y_0}{3!} \left(\frac{3^4}{4} - 3^3 + 3^2 \right) \right] \\ &= h \left[3y_0 + \frac{9}{2}(y_1 - y_0) + \frac{9}{4}(y_2 - 2y_1 + y_0) + \frac{3}{8}(y_3 - 3y_2 + 3y_1 - y_0) \right] \\ &= \frac{3h}{8} [y_0 + 3y_1 + 3y_2 + y_3] \end{aligned}$$

Composite Simpson 3/8 Rule

$$\begin{aligned} \bullet \quad I &= \int_{a=x_0}^{b=x_n} f(x)dx = \int_{x_0}^{x_3} f(x)dx + \int_{x_3}^{x_6} f(x)dx + \int_{x_6}^{x_9} f(x)dx + \cdots + \int_{x_{n-3}}^{x_n} f(x)dx \\ &= \frac{3h}{8}(y_0 + 3y_1 + 3y_2 + y_3 + y_6) + \frac{3h}{8}(y_3 + 3y_4 + 3y_5 + y_6 + y_9) + \cdots + \frac{3h}{8}(y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n) \\ &= \frac{3h}{8} \left[y_0 + y_n + 2 \sum_{i=1, i\%3=0}^{n-1} y_i + 3 \sum_{i=1, i\%3 \neq 0}^{n-1} y_i \right] \end{aligned}$$

Sample Problem: Evaluate - $\int_0^1 e^x dx$

Simpson's 3/8 Rule (n=3)

- Step size: $h = \frac{1 - 0}{3} = \frac{1}{3}$
- Divide the interval $[0, 1]$ into 3 equal subintervals: $x_0 = 0$, $x_1 = \frac{1}{3}$, $x_2 = 1\frac{2}{3}$, $x_3 = 1$
- Apply Simpson's 3/8 formula:

$$\int_0^1 e^x dx \approx \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$$

$$= \frac{3 \times \frac{1}{3}}{8} \left[e^0 + 3e^{\frac{1}{3}} + 3e^{\frac{2}{3}} + e^1 \right] = \frac{1}{8} \left[1 + 3e^{\frac{1}{3}} + 3e^{\frac{2}{3}} + e \right]$$

$$\int_0^1 e^x dx \approx \frac{1}{8} [1 + 3(1.3956) + 3(1.9477) + 2.7183] = \frac{1}{8} [1 + 4.1868 + 5.8431 + 2.7183] = \frac{1}{8}(13.7482) = 1.71853$$

Note on Choosing n for Trapezoidal, Simpson's $1/3$, and Simpson's $3/8$ Rules

- **Trapezoidal Rule:** n can be any +ve integer.
- **Simpson's $1/3$ Rule:** n must be **even** (divisible by 2).
- **Simpson's $3/8$ Rule:** n must be a **multiple of 3**.
- To satisfy all at once, choose n to be **multiple of 6** (the least common multiple of 2 and 3):
 $n = 6, 12, 18, \dots$

Gauss Quadrature

Why Gauss Quadrature?

- Uses non-equally spaced points (nodes)
- Integrates polynomials of degree up to $2n-1$ exactly with n points
- Interval is typically $[-1, 1]$ and transformed for general interval
- It also approximates integration as:

$$\int_a^b f(x)dx \approx \sum_{i=1}^n w_i f(x_i)$$

where x_i are carefully chosen nodes inside $[a, b]$ and w_i are weights.

Gauss 2-Point Quadrature

Exact integral up to degree 3 ($\because 2n-1 = 2*2 - 1 = 3$)

- $I = \sum_{i=1}^2 w_i f(x_i) = w_1 f(x_1) + w_2 f(x_2) = \int_{-1}^1 f(x) dx$
- Let, $f(x) = 1, \quad x, \quad x^2, \quad x^3$
- For $f(x) = 1$;
 $\int_{-1}^1 1 dx = w_1 + w_2 (\because f(x_1) = f(x_2) = 1) \Rightarrow w_1 + w_2 = 2 \dots (i)$

Gauss 2-Point Quadrature(Contd...)

Exact integral up to degree 3 ($\because 2n-1 = 2*2 - 1 = 3$)

- For $f(x) = x$

$$\int_{-1}^1 x dx = w_1 x_1 + w_2 x_2 \quad \Rightarrow w_1 x_1 + w_2 x_2 = 0 \dots (ii)$$

- For $f(x) = x^2$

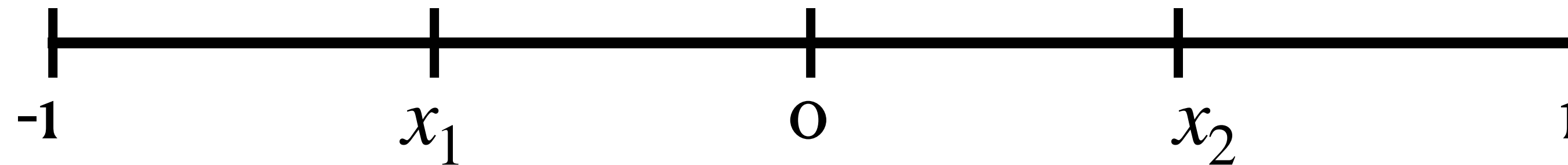
$$\int_{-1}^1 x^2 dx = w_1 x_1^2 + w_2 x_2^2 \quad \Rightarrow w_1 x_1^2 + w_2 x_2^2 = \frac{2}{3} \dots (iii)$$

- For $f(x) = x^3$

$$\int_{-1}^1 x^3 dx = w_1 x_1^3 + w_2 x_2^3 \quad \Rightarrow w_1 x_1^3 + w_2 x_2^3 = 0 \dots (iv)$$

Gauss 2-Point Quadrature(Contd...)

Exact integral up to degree 3 ($\because 2n-1 = 2*2 - 1 = 3$)



- So, $x_1 = -x_2$
- On solving;

$$w_1 = w_2 = \frac{1}{2}$$

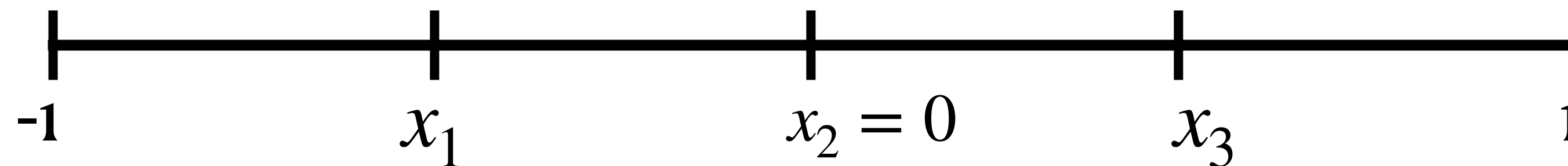
$$x_1 = -\frac{1}{\sqrt{3}}, \quad x_2 = \frac{1}{\sqrt{3}}$$

Gauss 3-Point Quadrature

Exact integral up to degree 5 ($\because 2n-1 = 2*3 - 1 = 5$)

- $I = \sum_{i=1}^3 w_i f(x_i) = w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3) = \int_{-1}^1 f(x) dx$

- Let, $f(x) = 1, x, x^2, x^3, x^4$



- So, $x_2 = 0, x_1 = -x_3$
- On solving

Gauss 3-Point Quadrature(Contd...)

Exact integral up to degree 5 ($\because 2n-1 = 2*3 - 1 = 5$)

- $w_1 = w_3 = \frac{5}{9}$
- $w_2 = \frac{8}{9}$
- $x_1 = -\sqrt{\frac{3}{5}}$
- $x_2 = 0$
- $x_3 = \sqrt{\frac{3}{5}}$

Sample Problem: Evaluate - $\int_{-1}^1 e^{-x^2} dx$
Using Gauss 2-point Quadrature

- Two Point Formula:

$$I = \int_{-1}^1 f(x) dx = \int_{-1}^1 e^{-x^2} dx = w_1 f(x_1) + w_2 f(x_2)$$

- We know by for two point formula:

$$w_1 = w_2 = \frac{1}{2}$$
$$x_1 = -\frac{1}{\sqrt{3}}, \quad x_2 = \frac{1}{\sqrt{3}}$$

- So, $I = 1 \cdot f\left(-\frac{1}{\sqrt{3}}\right) + 1 \cdot f\left(\frac{1}{\sqrt{3}}\right) = e^{\frac{-1}{3}} + e^{\frac{-1}{3}} = 2 \cdot e^{\frac{-1}{3}} = 1.43306$

Sample Problem: Evaluate - $\int_{-1}^1 e^{-x^2} dx$
Using Gauss 3-point Quadrature

- Three Point Formula:

$$I = \int_{-1}^1 f(x) dx = \int_{-1}^1 e^{-x^2} dx = w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3)$$

- We know by for three point formula:

$$w_1 = w_3 = \frac{5}{9}, \quad w_2 = \frac{8}{9}$$

$$x_1 = -\sqrt{\frac{3}{5}}, \quad x_2 = 0, \quad x_3 = \sqrt{\frac{3}{5}}$$

- So, $\frac{5}{9} \cdot e^{-\left(-\sqrt{\frac{3}{5}}\right)^2} + \frac{5}{9} \cdot e^{-0^2} + \frac{5}{9} \cdot e^{-\left(\sqrt{\frac{3}{5}}\right)^2} = 1.49868$

Mapping co-ordinates from $[a, b]$ to $[-1, 1]$

- Example: $\int_1^2 \frac{1}{x} dx$How will we apply Gauss 2 or 3-point quadrature formula here?

So, here comes the mapping the given limits into $[-1, 1]$.

- Let, $x = Az + B \Rightarrow dx = Adz$

- To find A and B

if $x = a$ (lower limit) and $z = -1$:

$$a = A \cdot (-1) + B \Rightarrow B - A = a \cdots (i)$$

and, if $x=b$, $z=1$:

$$A + B = b \cdots (ii)$$

Mapping co-ordinates from $[a, b]$ to $[-1, 1]$ (Contd...)

- Solving (i) and (ii):

$$B = \frac{b + a}{2}$$

$$A = \frac{b - a}{2}; \text{ Then,}$$

$$I = \int_a^b f(x)dx = A \int_{-1}^1 g(z)dz$$

Mapping co-ordinates from [a, b] to [-1, 1](Contd..)

- Example: $\int_1^2 \frac{1}{x} dx$

$$f(x) = \frac{1}{x}, a = 1, b = 2$$

- We know:

$$I = \int_a^b f(x) dx = A \int_{-1}^1 g(z) dz$$

$$A = \frac{b-a}{2} = \frac{2-1}{2} = \frac{1}{2}, \quad B = \frac{b+a}{2} = \frac{3}{2}, \quad \therefore x = Az + B = \frac{1}{2} \cdot z + \frac{3}{2} = \frac{1}{2} \cdot (z + 3)$$

- So, $I = \frac{1}{2} \int_{-1}^1 g(z) dz = \frac{1}{2} \int_{-1}^1 \frac{2}{z+3} dz = \int_{-1}^1 \frac{1}{z+3} dz \left(\because f(x) = \frac{1}{x} \Rightarrow g(z) = \frac{1}{\frac{1}{2}(z+3)} = \frac{2}{z+3} \right)$ now we can easily use the gaussian 2 or 3-point quadrature formula.

Any Questions?

Thank You